

symbolicsubstitution

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This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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1 Symbolic substitution

```
[1]: import casadi as ca
```

Let's build a trivial symbolic SX graph

```
[2]: x = ca.SX.sym("x")
     y = ca.SX.sym("y")
     z_ = x*y
     z = z_+x
     print(type(z), z)
```

```
<class 'casadi.casadi.SX'> ((x*y)+x)
```

We need SXFunction to manipulate the SX graph

```
[3]: f = ca.Function('f', [ca.vertcat(x,y)], [z])
```

We can substitute a leaf in the graph

```
[4]: w = ca.SX.sym("w")
     q = f(ca.vertcat(w,y))
```

f.eval() returns a tuple with all outputs, we selected the first

```
[5]: print(type(q), q)
```

```
<class 'casadi.casadi.SX'> ((w*y)+w)
```

Note how q is now an SX

We can take a shortcut via substitute:

```
[6]: q = ca.substitute(z,x,w)
     print(type(q), q)
```

```
<class 'casadi.casadi.SX'> ((w*y)+w)
```

Note that substitution of non-symbolic SX nodes is not permitted: substitute([z],[z_],[w]) This would throw an error

This is actually a restriction of Function: Function([[z_,y]], [z]) This would throw an error